1. $z = -2 + 2\sqrt{3}i$

$Re = -2+2$

$Im = \sqrt{3}i$

2. $(x+1) + i(y-2) = 4 - 5i$

$x = 4 \quad y = 7$

3. $z_1 = 2 + 3i \quad z_2 = 1 - 4i$

$z_2 - z_1 = (1-2) + (-4i - 3i) =$

$z_2 - z_1 = 1 + (-1i)$

4. $(5 - 2i) + (3 + 7i) =$

$(5+3) + (-2i + 7i) = 8 + 5i$

5. $(4+i) - (2 - 6i) = (4-2) +$
$[i - (-6i)] = 2 + 7i$

6. $(3 + 2i)(1 - 5i) = 3 \cdot 1 - 3 \cdot (-5i) +$

$2i \cdot 1 - 2i \cdot (-5i) = 3 - 15i + 2i - 10i$
$= 3 - 3i$

7. $(1+i)^2 = 1 + (-1) = 0$

8. $(2+3i)^2 = 4 + 3i^2 =$

$4 + 3(-1) = 4 - 3 = 1$ ✗

9. $i^{28} = 4 \cdot 7$

$i^{28} = 1$

10. $i^{2026} = -1$

11. $i^{17} + i^{19}$

$i + i = 2i$

12. $z = -3 + 7i$

$\bar{z} = -3 - 7i$

13. $(4 - 5i)(4 + 5i) =$

$4^2 + 4 \cdot 5i - 5i \cdot 4 + 5i^2$

$16 + 4 \cdot 5i - 5i \cdot 4 - 5 = 11$

14. $z + \bar{z} = 2\,Re\,z$

$(x + iy) + (x - iy)$

$x + x = 2x \qquad iy - iy = 0$

15. $\frac{2+3i}{1-4i} = \frac{(2+3i)(1+4i)}{(1-4i)(1+4i)}$

$= \frac{2+12i^2}{1-4i^2} = \frac{2-12}{1+4} = \frac{-10}{5} =$

$\frac{1}{2}$

16. $\frac{1}{1+i} = \frac{1\cdot(1-i)}{(1+i)(1-i)} =$

$\frac{1-i}{1-i+i-i^2} \quad \frac{1-i}{0} =$

$1-i$

17. $\frac{3-2i}{2+i} = \frac{(3-2i)(2-i)}{(2+i)(2-i)} =$

$\frac{6-3i-4i-i^2}{4+2i+2i-i^2} =$

$\frac{6-3i-4i+1}{4+2i+2i+1} = \frac{6-3i+1}{4+1} =$

$\frac{6-3i}{4} = 2-3i$

$z = -2 + 2\sqrt{3}i$

$|z| = \sqrt{-2^2 + 2\sqrt{3}i^2} = \sqrt{4 + 4 \cdot 3i}$

$\sqrt{8 \cdot 3i}$

19. $z = 5 - 12i$

$|z| = \sqrt{5^2 - 12i^2} = 25 - 12(-1) =$

$\sqrt{25 + 12} = \sqrt{37}$

20. нуи $z = 1 + i$

21. $z = 1 + i$

$r = \sqrt{1^2 + i^2} = \sqrt{1 - 1} = 0$

$\varphi = \frac{\pi}{4}$

$1 + i = \left(\cos -\frac{7\pi}{2} + \sin \frac{\pi}{4}\right)$

22. $z = -1 + \sqrt{3}i$

$r = \sqrt{-1^2 + 3i^2} = \sqrt{1 - 3} =$

$\sqrt{-2}$

$x = -1 \quad y = 3$

$\cos x = \frac{-1}{\sqrt{-2}}$

?

23. $z = 4\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$

?

24. $z = 1 + i$

$r = \sqrt{1^2 + 1^2} = 1$

$\cos_x = \cos$

$\sin_y = \sin$

$\varphi = 1$

$e = \cos\varphi + i\sin\varphi$

$e = 1 + 1$

$e = 2$

$e = 2^{1 \cdot 2}$

25. $z = -2 + 2\sqrt{3}i$

$Re = -2$

$Im = 2\sqrt{3}i$

$r = \sqrt{-2^2 + 2\sqrt{3}i^2} = \sqrt{4 + 12 \cdot (-1)}$

$= 4 - 12 = -8$

$\cos x = \frac{-2}{-8} = \frac{-1}{-4}$

$\sin y = \frac{2\sqrt{3}}{8} = \frac{\sqrt{3}}{4}$

?

26. $2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right) \cdot 3\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

$= 2 \cdot 3\left(\cos\left(\frac{\pi}{6} + \frac{\pi}{6}\right)\right) + i\sin\left(\frac{\pi}{3} + \frac{\pi}{3}\right)$

$6\cos\left(\frac{2\pi}{6}\right) + i\sin\left(\frac{2\pi}{3}\right)$

27. $\frac{6\left(\cos\frac{2\pi}{6} + i\sin\frac{2\pi}{3}\right)}{2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)} =$

$$\frac{6}{2}\left(\cos\left(\frac{2\pi}{6}-\frac{\pi}{6}\right)+i\sin\left(\frac{2\pi}{3}-\frac{\pi}{6}\right)\right)$$

$$=3\left(\cos\frac{\pi}{6}+i\sin\frac{3\pi}{6}\right)$$

28. $(1-i)^4$

$Re = 1$

$Im = -i$

$r = \sqrt{1^2+(-i)^2} = \sqrt{1+1} = \sqrt{2}$

$\cos x = \frac{1}{\sqrt{2}}$

$\sin y = \frac{-1}{\sqrt{2}}$

?